RODRIGO GOÑI

AI & Robotics Engineer

SUMMARY

AI Engineer with proven experience in the full lifecycle of on-premise Large Language Models (LLMs). Specialist in the fine-tuning (SFT, DPO) of models like Llama 3 and the development of APIs for deployment on GCP. Architected and developed a specialized chatbot, improving system resilience and fault management through a LangChain-based architecture.

CONTACT

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LANGUAGES

English: Level B2+ (Currently working North American company).
French: Level B1.
Spanish: Native.

SOLIDARITY PROJECTS

Programando el Futuro" is an initiative that provided an introductory course in programming using Scratch to eight enthusiastic young learners. This project was hosted at Escuela Pestalozzi 1-413, aiming to spark interest in technology and empower students by introducing them to the basics of programming. Our goal was to equip these bright minds with the tools to imagine and create their own digital projects, opening doors to future opportunities in technology.

EDUCATION

- **Postgraduate Degree in Artificial Intelligence | University of Buenos Aires (UBA)**
  - (Currently Enrolled)
  - Key Areas of Study: Machine Learning & Deep Learning, Computer Vision I & II, Natural Language Processing, Data Analysis, Machine Learning Operations (MLOps).
- **Mechatronic Engineering | Universidad Nacional de Cuyo (UNCuyo)**
  - March 2016 – June 2022
- **Honors & Awards:**
  - Distinguished Academic Recognition Diploma (Graduated 4th in class).
  - First Provincial Flag Escort (Awarded for 7th highest GPA across all engineering programs, 2016 cohort).
  - Final GPA: 8.80/10.
- **Thesis Project:**
  - DOF Robo-Fumigator (600 hours)Co-led the development of a ROS-based simulation for an autonomous mobile robot with dual SCARA arms.
  - Engineered a computer vision system using Deep Learning for weed detection and a depth camera for precise localization.
  - Implemented a Kalman filter to fuse GPS, accelerometer, and odometry data for robust navigation and geolocation of targets.
- **Electromechanical Technician | Ejército de los Andes ENET**
  - March 2010 – December 2015

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Artificial Intelligence Applied to Electrical Engineering (30 hours) | Institute of Electrical Energy (IEE), UNSJ - CONICETCompleted: Dec 2024, Grade: 10/10
- Introduction to 2D and 3D Quality Control by Artificial Vision | Universidad Nacional de Cuyo
- Automation by PLC | Siemens

SKILLS STACK AND TECNOLOGIES

PROGRAMMING SKILLS

- **Languages**:C, C++, Python, SQL, CUDA, Prolog, QML, Ladder, Structured Text, Bash Script, Docker, Flask, Dart.
- **Paradigms**: Object-Oriented, Functional, Scripting, Logical, Procedural.
- **Design Patterns**: Singleton, Decorator, Iterator, State.

SOFTWARE AND TOOLS

- **Operating Systems**: Linux (Debian, Ubuntu), Nvidia Jetpack, Windows Server 2019, Windows 11 Engineering
- **Software**: ROS1, ROS2, Gazebo, Simulink, Solid Edge, SolidWorks, KiCad, G-Code.

DEVELOPMENT ENVIRONMENTS

- **Frameworks & Libraries:** Matlab, AWS, Tensorflow, Keras, PyTorch, OpenMMLab, Nvidia TAO, DeepStream, TensorRT, Detectron2, OpenCV, GStreamer, LangChain , Hugging Face Transformers , Hugging Face Diffusers .
- **DevOps & Tools:** Git, GitHub Actions (CI/CD), Docker, Docker Compose, Kubernetes, Bitbucket, LLM Studio .

HARDWARE PROFICIENCY

- **Embedded Systems:** STM32 Discovery Board (ARM M4), Arduino (ATmega328), Jetson NANO (ARM A57), Jetson TX2, Jetson ORIN Nano and NX.

PROJECT MANAGEMENT

- **Methodologies:** SCRUM Tools: Trello, Jira.

SYSTEM MODELING AND CONTROL

- **Techniques:** State Space, PID Controllers, Fuzzy Logic Control.

COMMUNICATIONS PROGRAMMING

- **Protocols:** CAN, Serial, TCP/IP, UDP, Libraries: J1939, DDS, Socket TCP Programming.

MACHINE LEARNING AND AI

Core Techniques:

- **Regression & Classification:** Linear Regression.
- **Clustering:** k-means.
- **Probabilistic Models:** Bayesian Inference.
- **Search Algorithms:** Heuristic and state-space search techniques.
- **LLMs:** Llama models (Llama 2, Llama 3, Llama 3.1).
- **Vision Models:** U-Net, Mask R-CNN, ResNet, SIFT
- **Generative AI :** CogVideo, Wan2
- **Fine-Tuning Techniques:** SFT (Supervised Fine-Tuning), DPO (Direct Preference Optimization).

CLOUD PLATFORMS

- GCP, AWS, Firebase.

WORK EXPERIENCE

Student assistant

Universidad Nacional de cuyo | Agust 2019 - 2021

- Development and repair of study material for the General and Applied Electronics course.
- Coordination of student groups.

PPS Internships

Simen Rx | March 2021 - July 2021 (300 hours)

- Developed electronic boards using KiCad, focusing on high-efficiency design and reliability in deployment. Engaged in power electronics research, advancing knowledge in energy management and circuit optimization.
- Simulated the behavior of a high-power X-ray system in Simulink with MATLAB, incorporating various filters to mitigate voltage and current overloads. This simulation aimed to enhance system safety and performance under peak operating conditions.

Robot Operating System (ROS) Engineer

PLANTIUM.SA | July 2021 - October 2022

Computer Vision Project (OcuWeed) - (8-person team)

- Automated the Nvidia Jetpack embedded Linux installation process using Bash scripts, Linux libraries, CMake, and Make, significantly streamlining the setup.
- Achievement: Reduced the compilation and installation time from 25 minutes to 4 minutes, substantially increasing the work efficiency of the entire team.
- Developed the system structure using ROS, in C++ and Python for robust and flexible system functionality. Designed high-level system communications using ROS and handled low-level communications using CAN protocols to ensure reliable data transmission. Utilized Gazebo and rqt for system simulation, and implemented tf systems and GPS signals for accurate spatial geolocation of the robot.

Computer Vision AI Engineer

Ant-Automation | October 2022 - July 2024

Machine Vision Project (CIVA-CIRA) - Prevention and Maintenance using Machine Vision in Steel Mills (2-person team)

- Engineered an end-to-end thermal vision system to measure steel ladle freeboard, achieving 91% pixel detection accuracy which enabled distance inference to within tenths of a centimeter.
- Optimized the entire pipeline for high-throughput industrial use, achieving real-time performance of 20 FPS on a resource-constrained NVIDIA Jetson Nano by leveraging NVIDIA TensorRT for model inference and Numba for algorithm acceleration.
- Streamlined deployment and enhanced system reliability by automating the software installation process on embedded Linux platforms using Docker and Docker-Compose, ensuring consistent performance in a 24/7 operational environment.
- Developed and deployed multiple deep learning models for semantic segmentation and detection using a robust tech stack including PyTorch, TensorFlow, OpenCV, and the NVIDIA TAO ecosystem.
- Managed database operations and network communication using MySQL and Modbus TCP/IP. Achievement: Successfully developed a system that measures the freeboard level with pixel-level accuracy, significantly improving operational precision in harsh industrial.

Certification Project for FEMSA (Coca-Cola) - (7-person team)

- Developed a Python framework for data analysis and management of datasets in COCO format, optimizing the handling of complex data structures.
- Led AI model training, testing, and tuning using the Detectron2 framework, coupled with deploying the vision system structure through NVIDIA DeepStream SDK.
- Achievement: Enhanced the model's accuracy by 18%, increasing it from 65% to 83%. Additionally, by applying INT8 quantization, the inference time was dramatically reduced from 150 ms to 9 ms, enhancing system responsiveness.
- Set up a 4K streaming pipeline on Jetson NX Orin using UDP and GStreamer. Processed images with a U-Net mask to obtain Regions of Interest (ROI).

HotSpot Detection in Wheel Bearing Using Deep Learning Models - (1-person team)

- Engineered a high-precision thermal vision system for hotspot detection in wheel bearings, achieving 97% detection accuracy when validated against historical data.
- Implemented the core image processing pipeline using OpenCV and NumPy, enabling robust and reliable data analysis for the AI model.
- Managed and optimized the historical datasets with SQL to ensure high-quality data for model training and rigorous performance testing.

WORK EXPERIENCE

Software Engineer

Emtech SA | July 2024 - current

Developed an API based on Falcon with asynchronous behavior to facilitate communication with LLMs, specifically the Llama 3 model running in a cluster computing environment. Implemented key features including temperature control, socket communication for token streaming, and various functionalities to ensure OpenAI compliance. Currently working on an on-premise chatbot powered by the Llama 3 model.

Brand Alive Project

- Architected a custom, on-premise tool-use framework for smaller LLMs (Llama 3.1, Mistral), achieving ~90% precision in local function calls. This eliminated system-critical failures associated with external API dependencies and enhanced overall system robustness.
- Engineered a hybrid RAG and Supervised Fine-Tuning (SFT) pipeline that produced a conversational tone more authentically aligned with the author's brand than larger, general-purpose models like GPT-4.
- Developed and fine-tuned Llama 3.1 (8B) models on a specialized dataset from the author's literary works, proving the effectiveness of smaller, highly-specialized models for niche applications.
- Managed the full-stack deployment of the application, featuring a Python/LangChain backend on Google Cloud Platform (GCP) and a Dart-based frontend on Firebase.

GenVideo Project

- Engineered a sophisticated dataset processing pipeline that improved model training loss by 50% by integrating advanced filtering techniques, including Optical Flow, DOVER quality scoring, and Laplacian variance.
- Conducted a rigorous feasibility study on low-cost generative models (CogVideo, WAN2.), delivering a definitive, data-driven conclusion that prevented further investment in a non-viable technology path for the target application.
- Applied advanced fine-tuning methods (SFT, DPO) using the Hugging Face diffusers library to systematically evaluate and validate the performance limits of the models.

Football Analytics Project (current)

- Spearheaded the design and development of a computer vision platform for the intelligent analysis of football players from videos.
- Implemented state-of-the-art object detection and tracking algorithms to accurately monitor player and ball trajectories throughout the game.
- Engineered an AI-powered analytics engine to automatically process tracking data, enabling in-depth analysis of plays, pass patterns, and player movements.